HSU SYSTEM LIVE LINK

https://hsu-early-warning-system-student-retention-88dlnqbmp2cvxnnjcvy.streamlit.app/

Advisor Credintial:

Email: advisor@hsu.edu

Password: advisor123

Admin Credintial:

Email: admin@hsu.edu

Password: admin123

Student Credintials

Email	Name	Classification
uholloway@example.org	Amy Phillips	Freshman
woodmark@example.com	Anthony Harris	Junior
esnyder@example.com	Sharon Armstrong	Senior
wevans@example.org	Tyler Ortiz	Sophomore
wvaughn@example.com	Cody Rubio	Freshman
tholder@example.net	Jessica Smith	Sophomore
mercerregina@example.org	Paul Sanchez	Sophomore
clayrobert@example.net	Kara Clark	Junior
hernandezdavid@example.co	Diana Ford	Junior
clarkjesus@example.net	Christopher Bentley	Sophomore
twalker@example.net	Julian Cruz	Freshman
jenniferdurham@example.net	Jennifer Brown	Senior

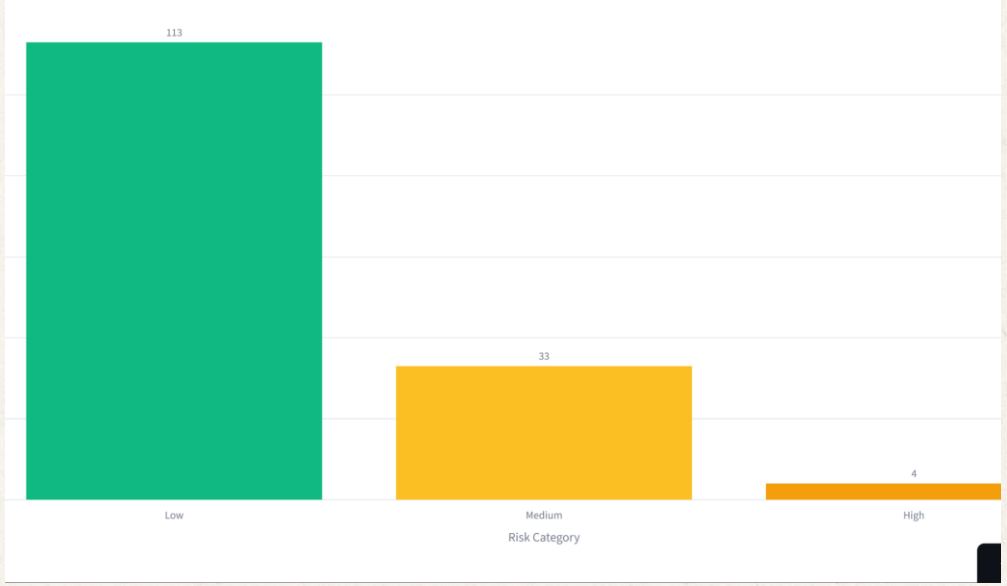
What your dashboard shows

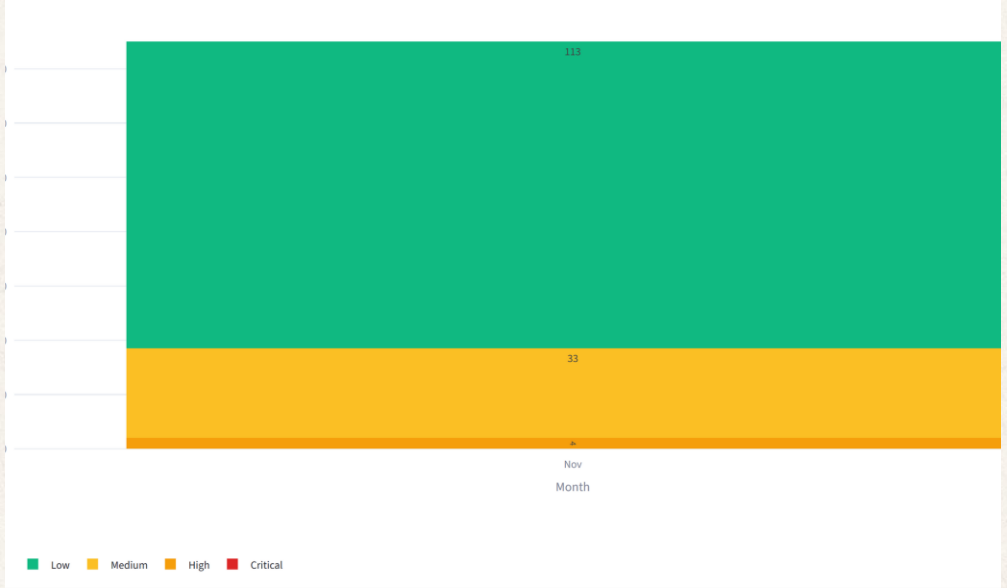
What we showed

- A **RiskCategory distribution chart** (Low / Medium / High).
- A **cohort chart** breaking risk down by **classification** (Freshman–Senior) and **first-generation status**.

What we observed from the visuals

- We concluded that the median students were in the Low-risk group and a significant number were in Medium-risk and so in the High-risk group. This ascertained that the model and data production did not categorize everyone as safe or risky, but a realistic blend created.
- When we cut risk by type, we also saw that the Freshmen and Sophomores were higher in Medium/High-risk students than Juniors and Seniors. This implied that the students in the initial stages of career were weaker and required more attention.
- After dividing by the first-generation status, we found that first- generation students were over-represented in the High-risk category, despite some of them having a fairly good high-school GPA. This was what we had anticipated, because non-academic barriers (engagement, finances, support) were viewed as very important by this group.
- One interesting trend though small was the fact that some students with reasonably high HS GPA ended up in Medium/High risk, particularly when they were poor in attendance and their use of LMS. That emphasized that past academic capability was not sufficient to maintain risk at a low level in case it would result in disengagement.





What insights can be drawn

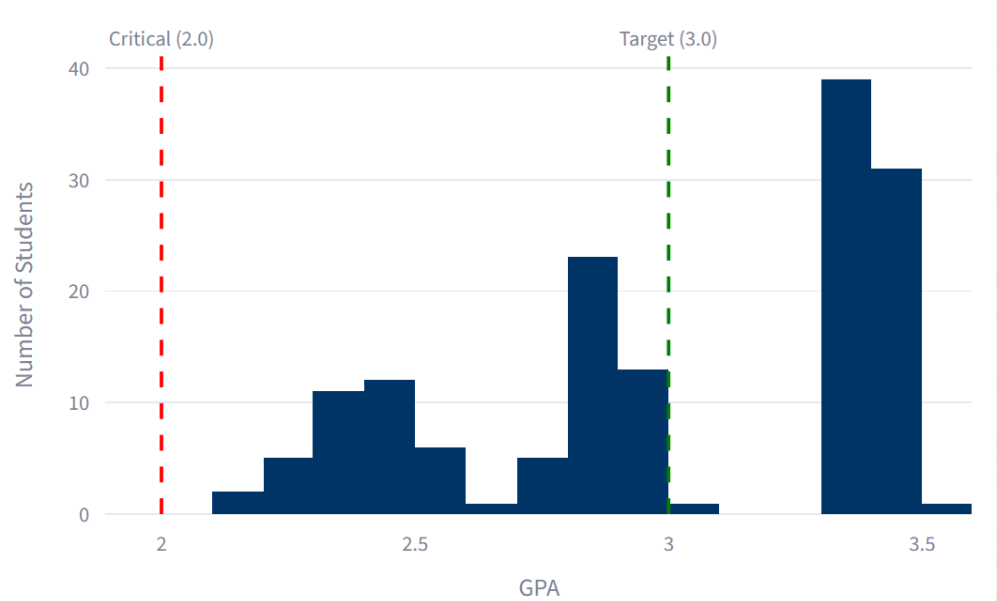
What we showed

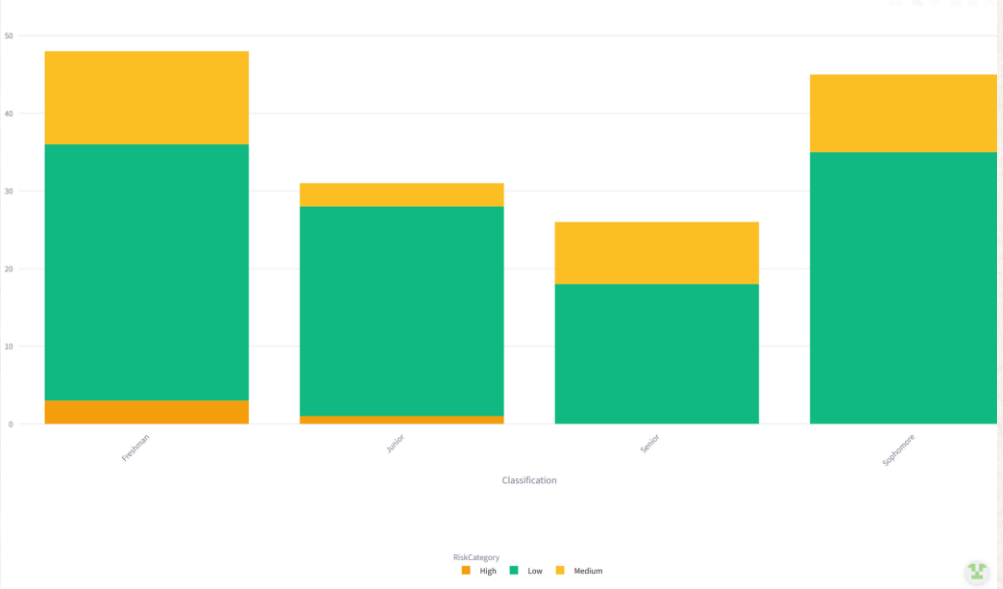
- Bar/box plots for **CalculatedGPA by RiskCategory**.
- Charts comparing **AttendanceRate** and **LoginsPerWeek** with risk level.
- Simple comparisons of **Balance / HasHold** and **NumCounselingVisits** across risk bands.

What we observed from the visuals

- We observed academic gradient: the means of CalculatedGPA decreased gradually as we advanced into the Low risk then to Medium risk then to High risk. The high-risk group of students was at the bottom of the GPA distribution, and this fact was consistent with the fact that academic difficulties are one of the primary causes.
- Student attendance and the number of LMS logins per week indicated that the presence of low attendance and extremely few logins per week as per the engagement charts resulted in the students being almost always placed in either the Medium or High-risk category. Put differently, there occurred a tendency of low engagement and high risk to manifest simultaneously.
- The financial charts showed that students who had high unpaid balances and active holds were significantly more frequent in the High-risk group. They were at a greater risk even in cases when their GPA was not dreadful due to a combination of financial stress and holds.
- The data on counseling revealed that students who had more than one visit or with the upper levels of severe rates were disproportionately High-risk, yet typically, a great number of High-risk students did not have a counseling record. What we interpreted this to mean was that a group of students who were in extreme distress already had some sort of life-line attached to them whereas the rest were out of sight and only captured through the academic and behavioral information..

This is what we concluded based on this slide: risk at HSU was multi-factorial: GPA and engagement showed the strongest predictive power, but financial and wellness indicators clearly divided the most risky patterns into very highest-risk groups.





How this supports decision-making

Who we felt should be contacted first

Tier 1 of priority Academic + Engagement crisis.

- High -risk students with GPA less than approximately 2.0-2.3, less than 80 attendance and extremely few LMS logins.
- These students were the ones we considered that, in case an advisor needs to contact or email a student, they appear to have red flags at the time.

Tier 2 of priority - Medium academic financial risk.

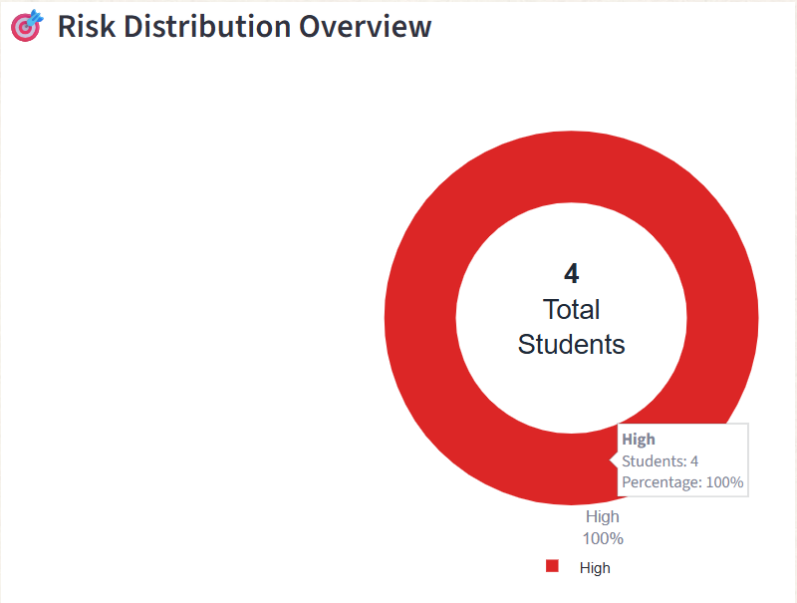
- Medium risk students with large balances, financial holds, or very low PaymentRate, particularly in the case of them having been first-generation as well.
- We sensed that this group should be offered active financial counseling and assistance outreach ahead of time, as any financial constraint would urge them into dropping out despite passing classes.

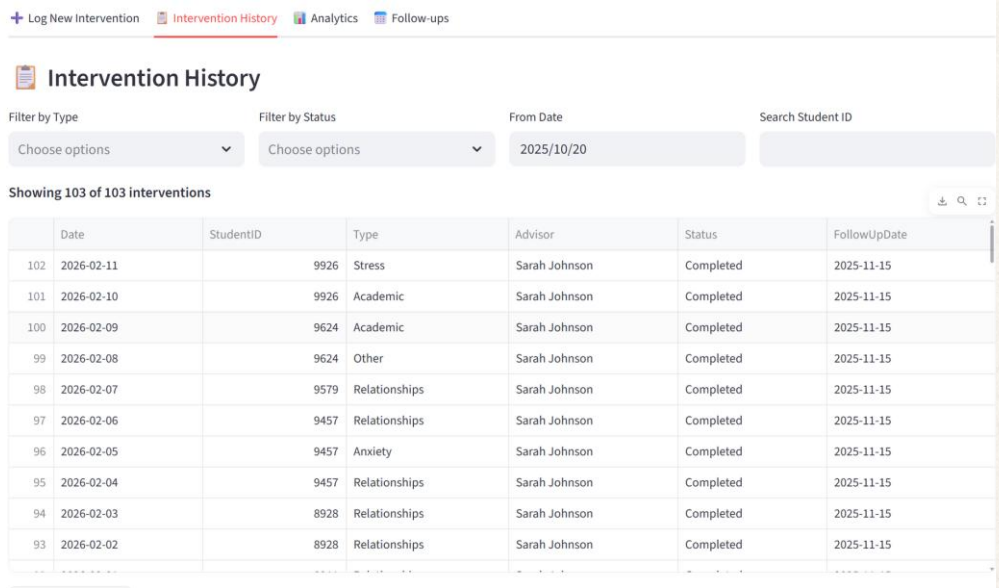
Priority Tier 3 Wellness and crisis indicators.

- Students who experienced repeated counseling visits or who had a high-severeity counseling visit and were increasing in their risk scores.
- In their case, we prescribed joint advisor and counseling services even where they had not lost GPA.
- What segments required special programs?

According to the cohort analytics, we recommended:

- First-generation freshmen and sophomores as a primary group to check-in during first semester, off-the-record study-skills training and mentorship.
- High balance and holds pupils with a high level as a group in order to conduct specific financial literacy lessons and payment plan messaging.





	Date	StudentID	Type	Advisor	Status	FollowUpDate
102	2026-02-11	9926	Stress	Sarah Johnson	Completed	2025-11-15
101	2026-02-10	9926	Academic	Sarah Johnson	Completed	2025-11-15
100	2026-02-09	9624	Academic	Sarah Johnson	Completed	2025-11-15
99	2026-02-08	9624	Other	Sarah Johnson	Completed	2025-11-15
98	2026-02-07	9579	Relationships	Sarah Johnson	Completed	2025-11-15
97	2026-02-06	9457	Relationships	Sarah Johnson	Completed	2025-11-15
96	2026-02-05	9457	Anxiety	Sarah Johnson	Completed	2025-11-15
95	2026-02-04	9457	Relationships	Sarah Johnson	Completed	2025-11-15
94	2026-02-03	8928	Relationships	Sarah Johnson	Completed	2025-11-15
93	2026-02-02	8928	Relationships	Sarah Johnson	Completed	2025-11-15